Cloud Architecture

Introduction to cloud databases

Summary Notes

Contents

Lesson objectives

By the end of this lesson, you should be able to:

- Define relational databases
- Explore database management
- Create Oracle APEX environment

What is a relational database?

A relational database is a type of database that organises data into one or more tables (or "relations") of columns and rows, with a unique identifier for each row. The columns represent attributes of the data, and the rows represent individual records. Relationships between different tables are established using common columns (keys) to link the data. This allows for data to be queried and manipulated in a structured and efficient way.

Characteristics of relational databases

A database table is a collection of data that is organised in a specific structure, with rows and columns. Each row represents a single record, and each column represents a field of the record. Tables are used to store and organise data. They can be used to store data of various types, such as text, numbers, and dates, and can be queried, updated, and manipulated using SQL (Structured Query Language). Data is usually constrained with the use of primary and foreign key which serve as unique identifiers.

Relational database management systems (RDBMS)

A RDBM is a software system that is used to manage and interact with a relational database. It allows users to create, read, update, and delete data stored in the database, as well as to define and manage the relationships between tables. RDBMS also provides tools for data integrity and security, such as enforcing constraints and managing user permissions.

Considerations for developing a RDBMS

- Content – It is important to take into account the type of data that will be stored in a database.
- Access – The access control to a database requires careful thought in that individuals require various degrees of access to the stored data.
- Physical organisation – The physical location and organisation is crucial for the architecture of an RDBMS, it impacts and array of factors such as system availability and back-up data capabilities.
- Logical structure – The logical structure of a RDBMS is based on the relational model, which is a way of organising data in a series of tables.

Role of a database administrator (DBA)

- Define the data needs of users
- Identify entities of interest
- Applies database programming languages
- Ensures data is secure from unauthorised access

Types of database management systems

- Flat files
- Single-user
- Multi-user

SQL database programming

SQL (Structured Query Language) is a programming language used for managing and manipulating relational databases. It is used for inserting, updating, querying and deleting data in a database. It can also be used for creating and modifying the structure of the database such as creating, altering, and dropping tables, views, and indexes. SQL statements are used to communicate with the database and are the standard language for relational database management systems.

Types of database management systems

- MySQL
- Oracle Database
- MS SQL Server